YEAR 7 · SCIENCE · MODELS · CHALLENGE HOMEWORK

Why Is Mr Bowen Still Awake?

Half-life with real coffee, a real interview, and carbon dating



Name: _____ **Class:** _____

Date given: _____ **Date due:** _____

How to do this homework

- **Parts A and B:** Mr Bowen's coffee. They teach you the method.
- **Part C is the main task.** You interview an adult in your family. **Do the interview at the weekend.**
- **Part D:** carbon dating.
- You need a **calculator**, a **ruler**, and a **computer or phone**. Answer in **full sentences**.

Remember from the lesson

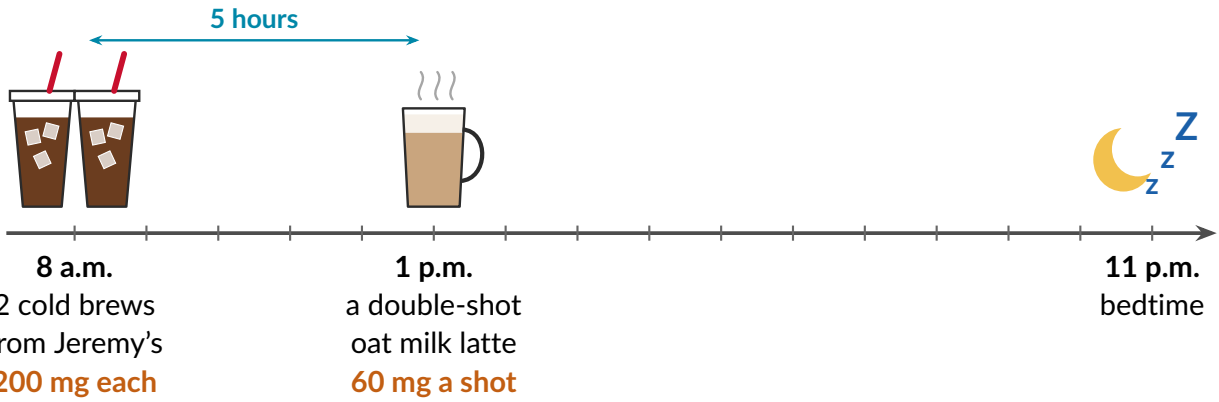
Half-life: the time it takes for half of something to disappear. Caffeine's half-life is about **5 hours**.

Carbon-14: a rare kind of carbon that slowly breaks down into nitrogen. Its half-life is **5 700 years**.

Carbon dating: finding the age of something that was once alive by measuring how much carbon-14 is left. It only works for things that were **alive**, up to about **50 000 years** old.

Part A Mr Bowen's Real Day

This is what Mr Bowen really drinks on a school day.



Word help

caffeine	a chemical in coffee, tea, cola and energy drinks. It keeps you awake.
mg	milligram. 1 000 mg = 1 g. One grain of rice is about 20 mg.
cold brew	strong coffee made slowly with cold water
double shot	two shots. A **shot** is a small cup of very strong coffee.
latte	coffee with a lot of milk. **Oat milk** is made from oats.

A1 The drinks

(a) The cold brews: $2 \times 200 =$ _____ mg (b) The latte: $2 \times 60 =$ _____ mg

(c) Does the **oat milk** add any caffeine? ☐ yes ☐ no

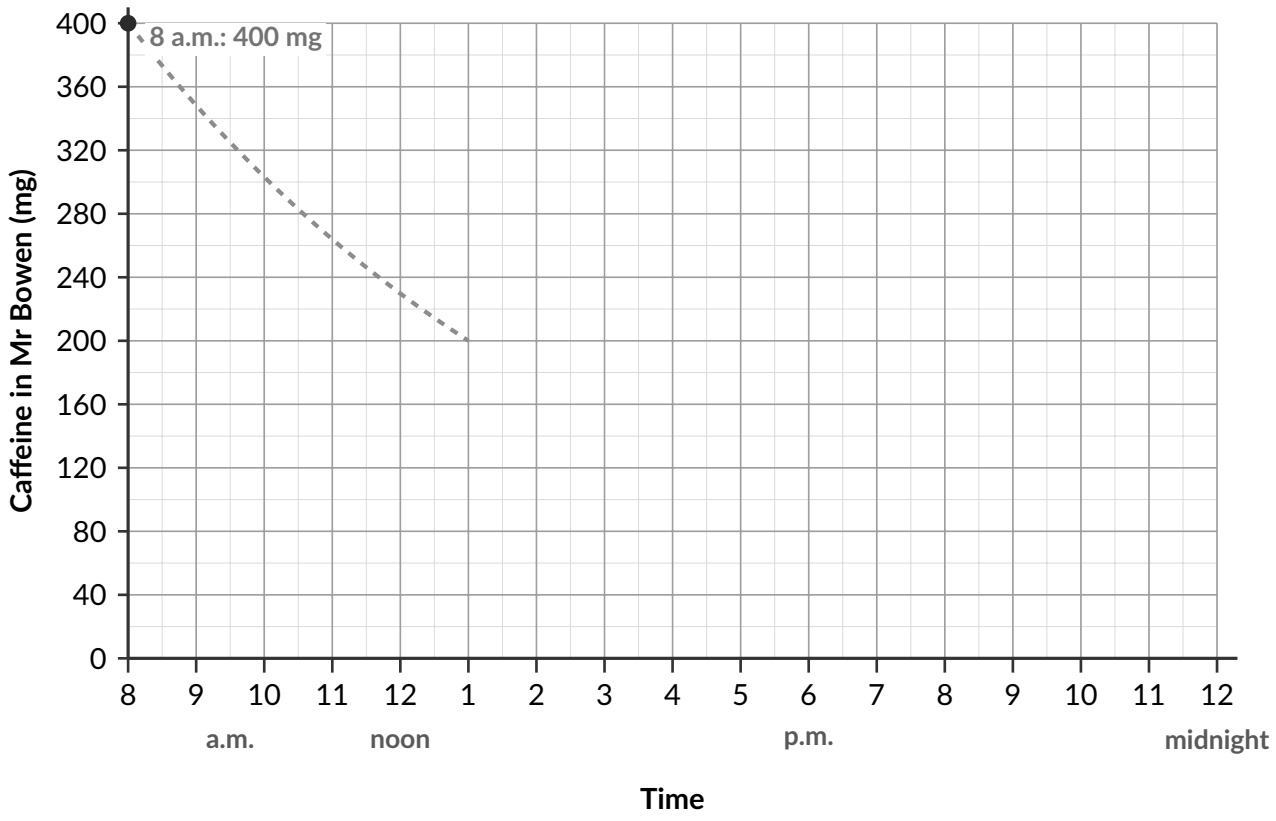
A2 Half-life steps

Fill in the boxes. Every 5 hours, **halve** (cut in half). When he drinks, **add**.

8 a.m. 400 mg → 5 hours ÷ 2 → **1 p.m.** _____ mg → latte + → **1 p.m.** after the latte _____ mg → 5 hours ÷ 2 → **6 p.m.** _____ mg → 5 hours ÷ 2 → **11 p.m.** bedtime _____ mg

A3 Draw the graph

Put a dot for every box in A2. The morning is drawn for you. At 1 p.m., draw a line **straight up**. Between the dots, draw a **smooth curve**.



A4 Read the graph

(a) At 8 p.m. Mr Bowen is marking your homework. Go **up** from 8 p.m. to your curve, then **across**. How much caffeine is in him? _____ mg

(b) How much caffeine is in him at bedtime, 11 p.m.? _____ mg

(c) Health experts say an adult should have **no more than 400 mg** of caffeine in a day. How much did Mr Bowen drink?

_____ + _____ = _____ mg Is that more than 400 mg? _____

(d) Tomorrow he wants **exactly 400 mg**. Which **one** drink should he not buy?

He should not buy _____.

A5 The title question

Why is Mr Bowen still awake at 11 p.m.? Use a number, and compare it with a cup of coffee (about 100 mg).

At 11 p.m. Mr Bowen still has …

Part B Hour by Hour

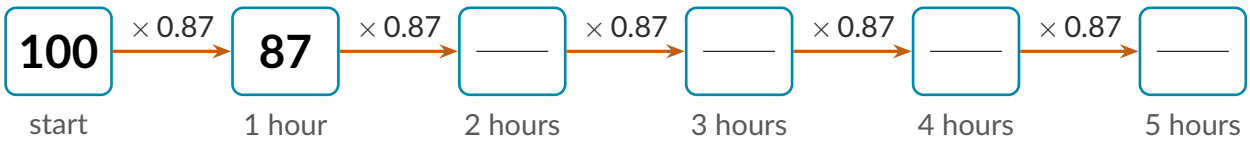
The hour-by-hour rule: it works for any time

Every hour: × 0.87. Round to a whole number. **When they drink: + the drink.**

Example: Mr Bowen at 1 p.m. 230 × 0.87 = 200, then + 120 for the latte = **320** mg.

B1 Why 0.87?

Start with 100 mg. Multiply by 0.87 every hour. Round each answer.



After 5 hours, about how much is left? _____ That is one _____.

B2 Check it with Mr Bowen

Use the rule from 1 p.m. to 8 p.m. (He drinks nothing in this time.)

1 p.m.	2 p.m.	3 p.m.	4 p.m.	5 p.m.	6 p.m.	7 p.m.	8 p.m.
320							

Is your 6 p.m. number about the same as in A2? ☐ yes ☐ no

Is your 8 p.m. number about the same as in A4(a)? ☐ yes ☐ no

When different methods agree, scientists _____ the answer.

Part C Interview a Caffeine Drinker

THE MAIN TASK

What you will do

Interview an adult in your family who drinks caffeine: a parent, a grandparent, an aunt, an uncle. Ask about coffee, tea, soda (like Coca-Cola) or **energy drinks like Sting**. Then **estimate** how much caffeine is in their body at bedtime.

Nobody at home drinks caffeine? Ask another adult you know. You can do the interview in **any language**. Write in **English**.

C1 The interview

Ask about **yesterday**, or another normal day.

1. "Can I interview you for my homework?"
2. "What did you drink that has caffeine?"
3. "What time did you drink it?"
4. "How big was it? A cup, a can, a bottle?"
5. "What brand was it?"
6. "What time did you go to bed?"

I interviewed my _____. **They went to bed at** _____.

Write their drinks here. Round each time to the **nearest hour**: 7:40 → 8:00, 10:20 → 10:00. Fill in the last two columns in C2.

Drink what, and how big	**Time**	**Rounded time**	**Caffeine (mg)**	**Where I found it**
Example: 1 can of cola, 330 ml	3:40 p.m.	4 p.m.	32	the company's website

C2 Find the caffeine

Where to look

1. **The label** on the can or bottle. Look for the word **caffeine**. It often says how many mg are in **100 ml**.
2. **The company's website.**
3. **Search:** *caffeine in [the drink] mg*, for example *caffeine in Sting 330 ml*.

Milk, sugar, ice and condensed milk: **0 mg**.

Two numbers, like 50 to 80 mg? Use the **middle**: 65 mg.



Reading a label. This bottle has 20 mg in every 100 ml. It holds 330 ml.

330 ÷ 100 = 3.3 3.3 × 20 = **66 mg** in the whole bottle.

(a) **10 mg in 100 ml**, a **330 ml** can: _____ × 10 = _____ mg

(b) **30 mg in 100 ml**, a **250 ml** can: _____ × 30 = _____ mg

Now look up every drink in C1. Fill in the **Caffeine** and **Where I found it** columns.

C3 Hour by hour

Use the rule from Part B and the **rounded** times. Row 1 is their **first** drink. Each row is **one hour later**. Stop at **bedtime**.

Start here

Time	Last hour × 0.87	+ Drink	Now (mg)

Then carry on here

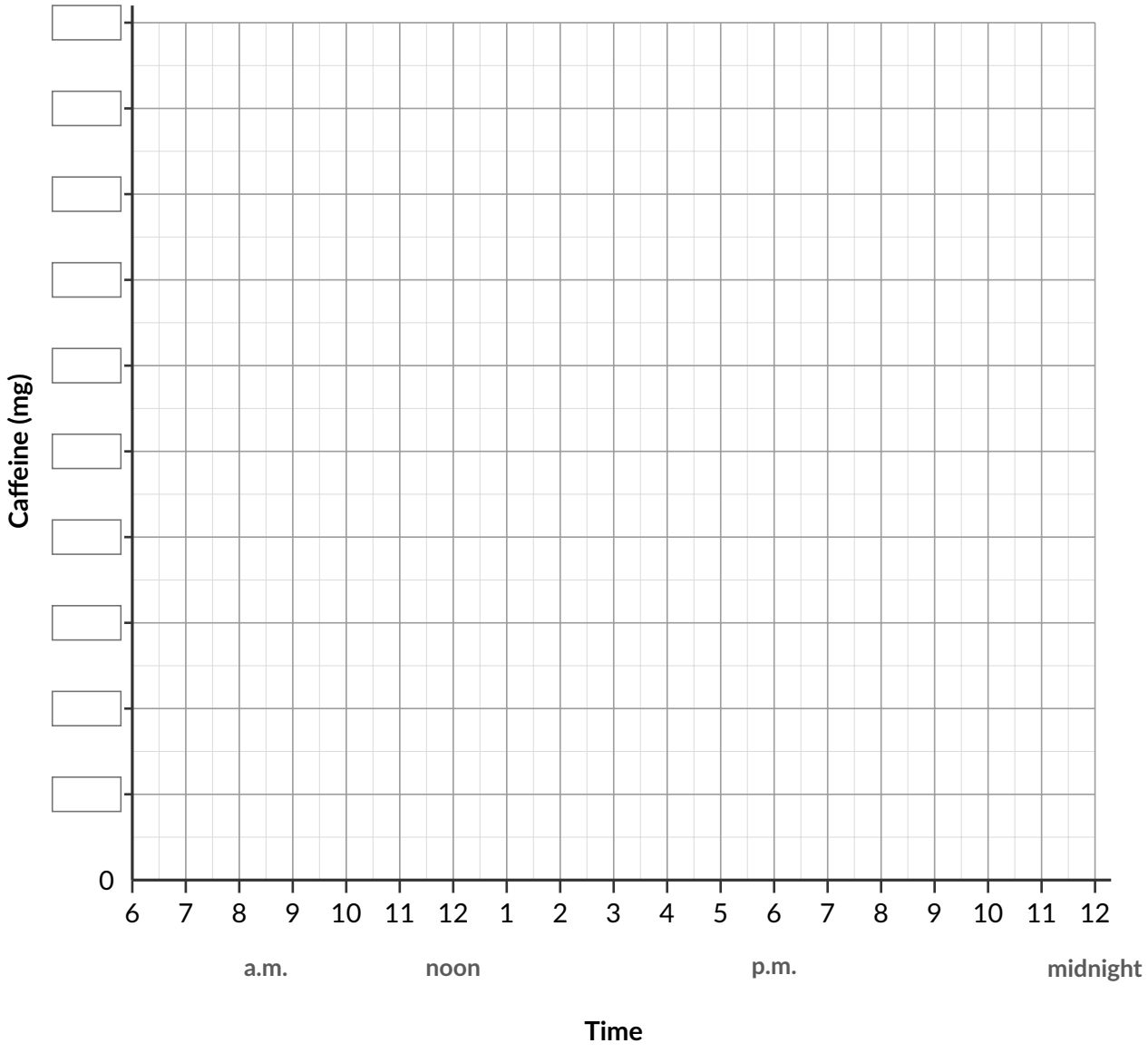
Time	Last hour × 0.87	+ Drink	Now (mg)

C4 Draw their graph

Step 1. Find the biggest number in your table. Choose one big square from this table, and write the numbers in the boxes on the graph.

Biggest number	under 100	under 200	under 300	under 400	under 600
One big square =	10 mg	20 mg	30 mg	40 mg	60 mg

Step 2. Put a dot for every row of your table. At a drink, go **straight up**. Between the dots, draw a **smooth curve**.



C5 Your results

(a) At bedtime (_____), my _____ had about _____ mg of caffeine in their body.

(b) Compare with Mr Bowen (80 mg at bedtime).

My _____ had more / less caffeine at bedtime than Mr Bowen, because …

(c) **Challenge.** They want less caffeine at bedtime. Give them **one** piece of advice. Use a number from your table.

I think they should … because …

(d) Your model is an estimate. Give **one** reason your bedtime number might be wrong. *Clue:* caffeine's half-life is not 5 hours for everybody. It can be from about 1.5 to 9.5 hours.

My bedtime number might be wrong because …

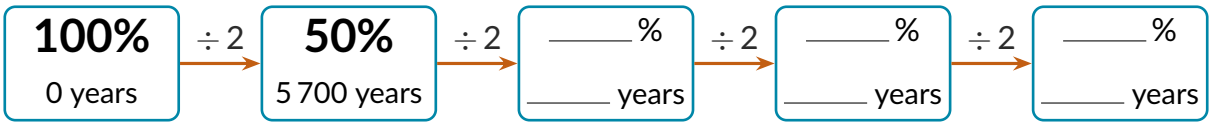
Before you hand it in: tick each one

- ☐ Every drink in C1 has a time, the caffeine in mg, and where I found it.
- ☐ My C3 table goes up **one hour** a row and stops at bedtime.
- ☐ Each drink is added **once**, in its own hour.
- ☐ My graph goes **straight up** at every drink.
- ☐ My bedtime number is the **same** in C3, C4 and C5(a).

Part D Carbon Dating

D1 The halving chain

Caffeine halves every 5 hours. Carbon-14 halves every **5 700 years**. Fill in the boxes.



D2 The museum

Scientists measured the carbon-14 left in six objects. Use D1. **Careful:** carbon dating does not work for everything.

Object	Was it ever alive?	Carbon-14 left	How many half-lives?	Age (years)
A wooden boat from a river	yes	50%	1	5 700
A bear bone from a cave		12.5%		
Charcoal (burnt wood)		6.25%		
A cotton T-shirt, made this year		100%		
A stone axe		—		
A *Tyrannosaurus rex* tooth		0%		

Word help: cotton comes from a plant · an **axe** is a tool for cutting wood

D3 Same model, different clock

While Mr Bowen is alive, **coffee** tops up his caffeine and **food** tops up his carbon-14. What is the **same** about caffeine and carbon-14? What is **different**? Use the word **half-life**.

They are the same because … The difference is …

Answer Key — Teacher Copy

Mr Bowen's numbers are chosen so Part A is exact: the drinks are 5 hours apart and bed is 10 hours after the latte. The rule in Part B rounds at every step and lands within 1 mg of Part A. Accept ±3 mg anywhere from a student who rounded at a different step.

A1 The drinks

(a) **400** mg · (b) **120** mg · (c) **no**

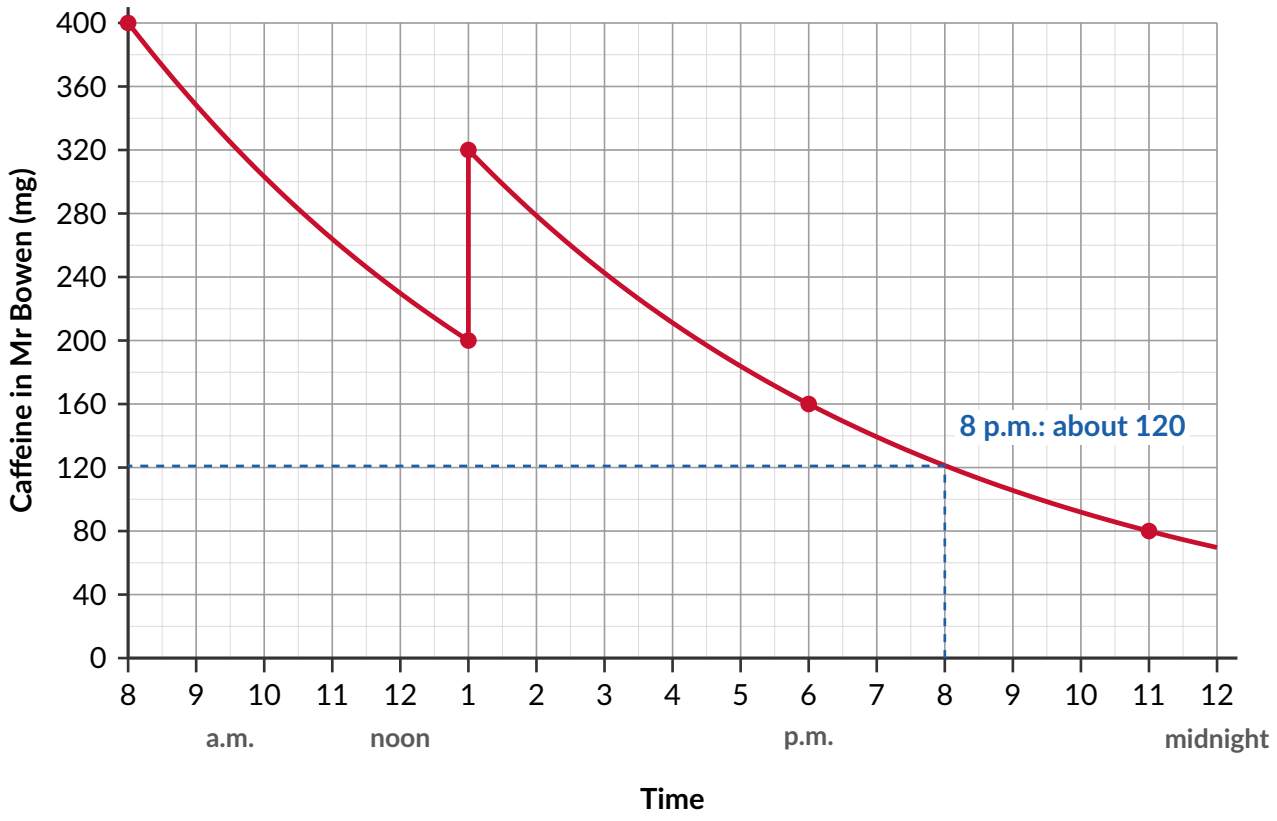
Expect 60 for (b) from a student who read *double-shot* as one shot. (c) is there for Part C: the condensed milk in a Vietnamese iced coffee adds nothing either, and a student who counts the whole drink will not know which number to look up.

A2 Half-life steps

1 p.m.: **200** · after the latte: **320** · 6 p.m.: **160** · 11 p.m.: **80**

The latte box is the question. Expect 100 and 50 from a student who halved straight through it, and 260 / 130 / 65 from one who added 60. If the second box is 520, they added the latte to the 8 a.m. number before halving — ask what happened between 8 a.m. and 1 p.m.

A3 The graph



Look for the **vertical** jump at 1 p.m. A slope from noon to 1 p.m. means the student is joining dots, not modelling a drink. Two straight lines instead of curves is the other common one; it makes A4(a) come out at 128 instead of 121.

A4 Read the graph

(a) about **120** mg (exactly 121; accept 115–130) · (b) **80** mg
(c) 400 + 120 = **520** mg; **yes** · (d) **the latte** (520 − 120 = 400)

(d): "one cold brew" gives 320, which is under 400 but not *exactly* 400 — worth a minute in class, because both are healthier and only one answers the question asked.

A5 The title question

At 11 p.m. Mr Bowen still has 80 mg of caffeine in his body. That is almost a whole cup of coffee.

Check for the number *and* the comparison. "Because he drank coffee" is true and is not the answer.

B1 Why 0.87?

100, 87, **76, 66, 57, 50** · about **50 mg**, which is **half**: one **half-life** ($0.87^5 = 0.498$)

B2 Check it with Mr Bowen

320, **278, 242, 211, 184, 160, 139, 121** · yes, yes · scientists **trust** the answer

The last line is deck slide 43, word for word. Expect 160, 80, 40, 20 … from a student who halved every hour instead of × 0.87 (both of B2's checks then come out "no"): that is the error to catch before Part C, where it would put every family member at almost zero by bedtime.

C1–C4 The interview, the table, the graph

No single answer. Mark in this order, and stop at the first thing that is missing — it is the thing to send back:

1. A real person, named by relationship ("my father"), with a bedtime.
2. Every drink has a time, a rounded time, an amount in mg, and **where the number came from**. "Google" is acceptable; a blank is not.
3. C2: (a) 3.3 × 10 = **33** mg · (b) 2.5 × 30 = **75** mg.
4. C3 starts at the first drink, goes up one hour a row (left table, then right), and stops at bedtime. Each row is the row above × 0.87; a drink appears *once*, in its own row.
5. C4: the scale goes up in equal steps from the table, the dots match C3, drinks are vertical jumps, and the bedtime value on the graph is the last row of C3.

Numbers to sanity-check against (all approximate; the point is that the student found one and said where): Sting, 330 ml bottle, about 50–85 mg (the Vietnamese label gives a range per litre, which is a good C2 conversation) · cola, 330 ml can, about 30–35 mg · Red Bull, 250 ml can, about 50–80 mg depending on the version · one espresso shot, about 60–75 mg · Vietnamese iced coffee, about 100–200 mg (robusta, and very variable) · a cup of black or green tea, about 30–50 mg.

Expect at least one sheet where bedtime is above Mr Bowen's 80 mg. An afternoon iced coffee and an evening Sting will do it. That sheet is worth showing the class (without the name).

C5 Results

(a) must match the last row of C3. (b) needs *because* and a reason from the table: more drinks, a bigger drink, or a later drink. (c) the strongest advice moves or cuts the **last** drink of the day — an afternoon drink matters far more at bedtime than a morning one. Credit a student who reworks the table to show it. (d) accept: the half-life is different for different people (the clue); the caffeine numbers are estimates and brands differ; times were rounded; the person may not remember exactly; the model adds a drink all at once, but caffeine takes time to get into the blood; yesterday may not have been a normal day. "I might have made a mistake" is not a limitation of the model — ask which part of the model.

D1 The halving chain

100%, 50%, **25%, 12.5%, 6.25%** · 0, 5 700, **11 400, 17 100, 22 800** years

D2 The museum

bear bone	yes	12.5%	3	**17 100**
charcoal	**yes** (it was wood)	6.25%	4	**22 800**
cotton T-shirt	**yes** (cotton is a plant)	100%	0	**about 0** (it is new)
stone axe	**no**	—	—	carbon dating does not work
T. rex tooth	yes	0%	—	too old: over 50 000 years

The three traps are the question. Expect "no" for charcoal and the T-shirt from students who read *alive* as *an animal*. Expect "no" for the *T. rex* because it is dead — it was alive; it is too old (deck slide 31: 66 million years, over 11 000 half-lives). A student who writes "date the volcanic ash" has remembered slide 34; give credit.

D3 Same model, different clock

They are the same because both get topped up while he is alive, and when the top-ups stop, both halve every half-life. The difference is the half-life: 5 hours for caffeine and 5 700 years for carbon-14.

Deck slide 25: "food and drink keep topping up your carbon-14. Even coffee." A student who says only "they both halve" has the first half; push for the difference in the clock.